MUSANZE HARVEST INTELLIGENCE SERVICE

SWE 3513 — Artificial Intelligence
Individual Assignment 2

Submission detail	Value
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Verification code	AI-A2-7790
Personalized seed	7790
Python version	3.14.7

Problem statement

This project demonstrates a machine-learning workflow for a synthetic potato-cooperative dataset. It estimates harvest yield, classifies high-risk conditions, and explores groups of records through clustering.

Data notice: The dataset is synthetically generated for educational purposes. The models have not been validated for real farming decisions.

1. Dataset and Data Preparation

The personalized data generator uses NumPy's seeded random generator with seed 7790. The dataset includes rainfall, average temperature, soil moisture, fertilizer quantity, field size, season, cooperative support, harvest yield, and a high-risk label.

Required schema

Feature	Description
rainfall_mm	Rainfall in millimetres
avg_temp_c	Average temperature in degrees Celsius
soil_moisture_pct	Soil moisture percentage
fertilizer_kg	Fertilizer quantity in kilograms
field_size_ha	Field size in hectares
season	Categorical season (A or B)
cooperative_support	Binary cooperative support indicator
yield_tons	Regression target: harvest yield in tons
high_risk	Classification target: risk indicator

Cleaning and quality checks

The generator produced 243 rows, including three deliberate exact duplicates and missing values in soil moisture and fertilizer. Cleaning removed the three duplicate rows and imputed missing values, resulting in 240 rows and nine columns with no missing values remaining.

The generator records a SHA-256 fingerprint of the generated CSV in artifacts/data_fingerprint.txt. The fingerprint helps verify whether the generated file's bytes remain unchanged.

Data preparation is part of the reproducibility process. Imputation, scaling, and categorical encoding used in model pipelines are fitted on training data rather than the held-out test set.

2. Modelling Methodology

Regression

The regression task predicts yield_tons. A mean-prediction baseline was compared with linear regression and random forest regression. Models were evaluated using mean absolute error (MAE), root mean squared error (RMSE), and coefficient of determination (R^2). Lower MAE and RMSE are better; R^2 describes the proportion of test-target variation explained by the model.

Classification

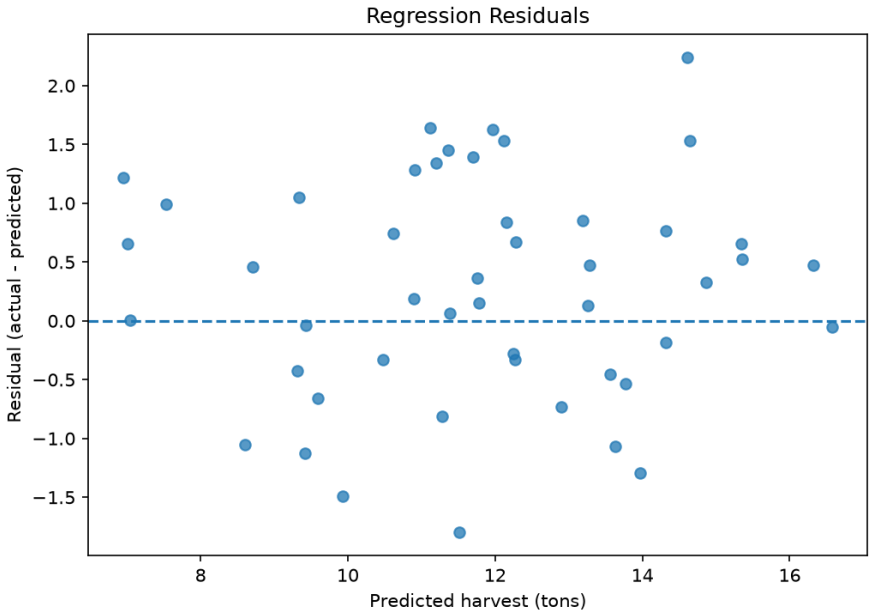
The classification task predicts high_risk. A majority-class baseline was compared with logistic regression and random forest classification. Precision, recall, F1, confusion matrices, and ROC-AUC were recorded where the metric was valid. The high_risk target was excluded from input features.

Clustering

K-means clustering used input features after preprocessing. Candidate cluster counts from K=2 through K=6 were evaluated using silhouette scores. A higher silhouette score generally indicates better separation, although it does not prove that clusters have real-world meaning.

Leakage prevention and evaluation

The regression target was not used as a predictor, and target-derived features were excluded. The cleaned data was divided into 192 training rows and 48 test rows using the configured random seed. The held-out test set was used for model comparison.



3. Regression Results

The selected regression model was linear regression because it achieved the lowest test RMSE among the compared models.

Model	mae	rmse	r2
metrics	N/A	N/A	N/A

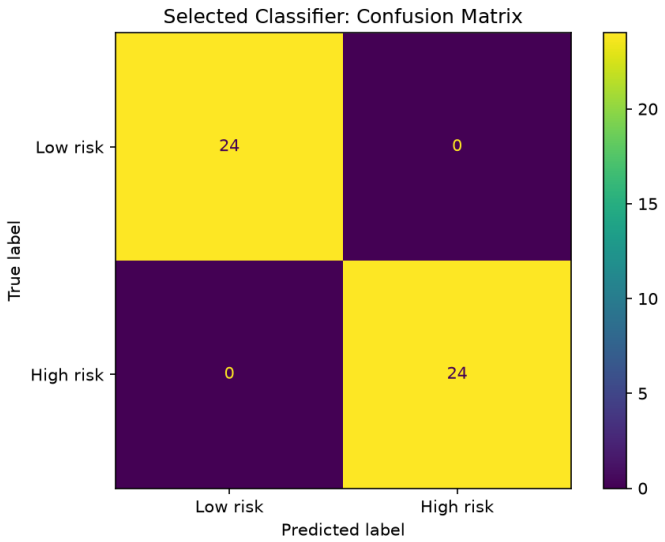
Interpretation

Linear regression achieved MAE of approximately 0.798 tons, RMSE of approximately 0.963 tons, and R^2 of approximately 0.864. It performed better than the mean baseline and the tested random forest regressor on this synthetic test split.

These metrics measure performance on the generated dataset only. They do not demonstrate accuracy on real harvests. RMSE penalizes larger errors more strongly than MAE, while R^2 can be negative when a model performs worse than a constant mean predictor.

Residual analysis

Residuals are the differences between observed and predicted yields. Their pattern can reveal systematic errors or unusually large mistakes. This plot should be interpreted alongside the numerical metrics.



4. Classification Results

The selected classifier was random forest. Its test-set precision, recall, F1, and ROC-AUC were each 1.000 on the recorded split.

Model	precision	recall	f1	roc_auc
metrics	N/A	N/A	N/A	N/A

Confusion matrix

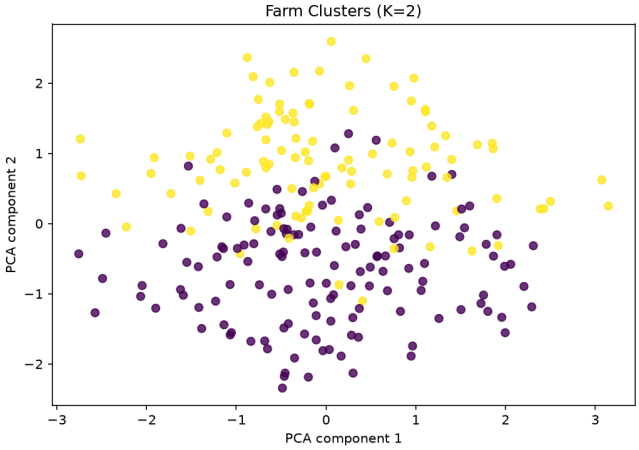
The confusion matrix compares predicted labels with actual test labels. False positives are high-risk predictions for records whose true label is negative; false negatives are missed high-risk records.

Important caution: Perfect test scores on a small synthetic dataset can result from the way the labels were generated. They are not evidence that the classifier will achieve perfect performance on actual farms.

5. Clustering, API, Testing and Responsible AI

Clustering results

The clustering report stores candidate silhouette scores and the selected cluster count. The highest observed score was approximately 0.162 at K=2.



K=2 was selected, with a silhouette score of approximately 0.162. This low score suggests that clusters overlap and are not strongly separated. Cluster profiles should be treated as exploratory summaries.

API and automated tests

The FastAPI service exposes GET /health, GET /model-info, and POST /predict. Input validation is intended to reject missing fields, incorrect types, extra fields, and out-of-range values. The latest recorded automated test run passed all six tests, including inference, validation, reproducibility, and API response checks.

Responsible use and limitations

This is an educational prototype using synthetic data, not a production agricultural service. It may not represent actual weather, soil, crop varieties, pests, local farming methods, or seasonal conditions. Do not use its output as the sole basis for loans, insurance, aid allocation, or other high-impact decisions. Real-world use would require representative field data, independent validation, agricultural expert review, privacy protections, and ongoing monitoring. See docs/RESPONSIBLE_AI.md.

Reproducibility: Run `python run_all.py --student-id "25/27790" --seed 7790` from the project root using the project's Python environment to regenerate the data, models, metrics, plots, and tests.